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PI Name:	Clement, Gilles Ph.D.		
Project Title:	Functional Task Tests in Partial Gravity during Parabolic Flight		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
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Space Biology Special Category:	None		
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Zip Code:	77058-3711	**Congressional District:**	36
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:	May 2023 report: Gilles R. Clément, PhD, Principal Investigator, KBR, NASA Johnson Space Center, Houston TX; Timothy Macaulay, PhD, Co-Investigator, KBR, NASA Johnson Space Center, Houston TX; Austin Bollinger, KBR, NASA Johnson Space Center, Houston TX; Scott Wood, PhD, Co-Investigator, NASA Johnson Space Center, Houston TX.		
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Performance Goal Text:

Task Description:

Critical mission tasks that are required by crews immediately after landing on a planetary surface include egressing from a seat, jumping, and walking. To define an effective and comprehensive countermeasure strategy for preserving crew performance during exploration-class missions, there is a need to understand how these functional tasks are performed in partial gravity, such as on the Moon or Mars.
We will analyze the execution of four critical mission tasks (Seat Egress and Walk, Recovery from Fall and Stand, Jump Down, and Tandem Stance) during the partial gravity and normal gravity phases of parabolic flight by using the same equipment and procedures as those previously used on astronauts returning from International Space Station (ISS) missions and ground-based subjects during axial body unloading. Our hypothesis is that the limits of stability for these activities will be larger when the gravity level is reduced. The largest decreases in performance are expected at the lowest gravity level (0.25 g) because subjects will no longer be able to use the gravitational reference for their perception of upright. Ultimately, this information could be used to assess performance risks and inform the design of countermeasures for NASA exploration-class human missions.

The four specific aims include:

Specific Aim 1: Seat Egress and Walk. The purpose of this test is to measure the ability to rise from a seated position and walk while avoiding obstacles to test mobility. This test is identical to the Sit-to-Stand and Walk-&-Turn test used for Standard Measures after spaceflight and bed rest. In this test, subjects are requested to rise from a seated position as quickly as possible without using their hands and walk as quickly and safely as possible straight ahead towards a cone (4 m distance), walk around the cone, then return and sit back down in the chair. On the way to and back from the cone, subjects step over a 30-cm obstacle. Two trials will be performed per parabola. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units. Performance metrics include time to complete the trial, turn rate during the turn, obstacle contact, and head-torso coordination.

Specific Aim 2: Tandem Stance. The Tandem Stance test is a standard test of static postural stability. This test is similar to the Computerize Dynamic Posturography test performed on astronauts as part of their Medical Requirements and on bed rest subjects as part of the HRP Standard Measures (Postural Equilibrium Control). In this test, at the sound of a tone subjects are instructed to stand upright in a heel-to-toe fashion with their arms crossed on their chest. This test is performed with the eyes open and with the eyes closed. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units. The maximum time (prior to taking a step) as well as the medial-lateral peak-to-peak sway angle (p-p sway) is used to quantify postural stability.

Specific Aim 3: Recovery from Fall and Stand. The purpose of this test is to measure the ability to maintain postural control after standing up from a prone position. Impairment in the ability to rise from a prone position is one of the strongest independent risk factors associated with serious fall-related injuries. In this test, subjects rest in a prone position, then stand up as quickly as possible and maintain a quiet standing position. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units. The anterior-posterior and medial-lateral peak-to-peak sway angle (p-p sway) is used to compute the equilibrium score, where 12.5 is the maximum theoretical p-p sway. This test also induces an orthostatic challenge. Therefore, heart rate is collected continuously throughout this test. This cardiovascular data is used to detect potential signs of orthostatic intolerance during this active head-up tilt test.

Specific Aim 4: Jump Down. In the Jump Down test, at the sound of a tone subjects perform a two-footed hop from a height of 30 cm onto a force plate that measures the ground reaction forces on landing. After landing, subjects are instructed to remain still on the force plate, in the standing position, with arms at their sides for 10 s. After 10s, subjects will also perform a maximal voluntary lean in one direction to quantify changes in the limits of stability at different g-levels. Two jump-down trials will be performed per parabola. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units.

Study Participants. Twelve subjects (6 male, 6 female) will be tested during 3 flights of 30 parabolas, including 10 parabolas at 0.25 g, 10 parabolas at 0.5 g, and 10 parabolas at 0.75 g. In addition, each subject will perform all the functional task tests in 1 g during the flight between parabolas when the aircraft flies straight and level.

Risk Characterization, Quantification\Evidence. This study will contribute to gap closure by providing information regarding any functional task performance deficits in partial gravity. The dose-response relationship between gravity level and task performance decrement will also help in determining the gravity threshold for these functional tasks. These functional task tests are selected to simulate critical mission tasks that crewmembers may be required to perform when they land on another planet with partial gravity.

Countermeasure\Prototype Hardware or Software. This task will contribute to gap closure by determining the gravity threshold for these functional tasks.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The functional tasks tests in the proposed study will challenge balance control, a function that is paramount to the efficient completion of critical mission tasks. The vestibular and sensorimotor systems play a fundamental role in balance control, and the functioning of these systems is altered during parabolic flight. Knowledge gained from the proposed study will allow us to characterize the risk of balance impairments in various partial gravity levels, thus ensuring that a more effective and comprehensive countermeasure strategy can be developed for preserving crew performance during exploration-class missions.

Task Progress:	Twelve subjects flew aboard a NOVESPACE Airbus A310 Zero-G aircraft on June 13-15, 2023, in Bordeaux, France. Each flight included 31 parabolas: 1 test parabola, 10 parabolas at 0.25 g, 10 parabolas at 0.5 g, and 10 parabolas at 0.75 g. Four subjects were tested each of three flights, i.e., a total of 12 subjects. The subjects were tested during parabolas and also performed tests in 1g between parabolas. Data were collected using inertial measurement units (Opal V2, APDM, Portland, OR) worn on the head and trunk, heart rate monitors (RS800CX, Polar, Kempele, Finland), and a force plate (Bertec, Columbus, OH). Because there were no opportunities to test on the ground, 14 other healthy subjects were tested using identical procedures in a 1 g laboratory setting. Gravity level had a significant effect on performance, with the greatest changes from 1 g tending to be at the 0.25 g level. Lower gravity levels were associated with (1) increased times to complete the seat egress and walk task and the recovery from fall task; (2) decreased change in heart rate during the recovery from fall task; (3) decreased rail balance times; and (4) increased cone of stability distance in the anterior-posterior direction. These preliminary results were presented at the 2024 NASA Human Research Program Investigators' Workshop and are reported in a manuscript that has been submitted to Nature Microgravity. A follow-up analysis of head motion, namely angular velocity around the yaw plane, confirmed the slower turn rates during the seat egress and walk task. In addition, lower gravity levels were associated with larger magnitudes of upward head pitch while exiting the turn compared to entering the turn, suggesting that subjects might have adjusted their gaze as a strategy for maintaining balance or overall functional performance in partial gravity. These data suggest that there is a negative dose-response relationship between gravity level and functional task performance. The largest changes in performance were expected at the lowest gravity level (0.25 g) because subjects would no longer be able to use the gravitational reference for the perception of upright. Understanding the extent of performance deficits informs the risks and design of countermeasures for exploration spaceflight missions.
Bibliography Type:	Description: (Last Updated: 10/10/2025)
Abstracts for Journals and Proceedings	Macaulay T, Bollinger A, Wood SJ, Reschke MF, Clément G. "Functional task tests in partial gravity during parabolic flight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 5-9, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 5-9, 2024. , Feb-2024